

class10

Tanieng Huang

Structural Bioinformatics (Pt. 1)

2. Introduction to the RCSB Protein Data Bank (PDB)

- **Q1:** What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
# Read the CSV file into R
stats <- read.csv("pdb_stats.csv", row.names = 1)

stats
```

	X.ray	EM	NMR	Integrative	Multiple.methods	Neutron
Protein (only)	178795	21825	12773	343	226	84
Protein/Oligosaccharide	10363	3564	34	8	11	1
Protein/NA	9106	6335	287	24	7	0
Nucleic acid (only)	3132	221	1566	3	15	3
Other	175	25	33	4	0	0
Oligosaccharide (only)	11	0	6	0	1	0

	Other	Total
Protein (only)	32	214078
Protein/Oligosaccharide	0	13981
Protein/NA	0	15759
Nucleic acid (only)	1	4941
Other	0	237
Oligosaccharide (only)	4	22

```
# Function to remove commas and sum a column
comma_sum <- function(x) {
  # Remove commas, convert to numeric, then sum
  sum(as.numeric(gsub(",", "", x)))
}
```

```
# Calculate totals for each method across all molecular types
xray_total <- sum(stats$X.ray)
em_total <- sum(stats$EM)
overall_total <- sum(stats$Total)

# Percentage for X-ray
xray_total / overall_total * 100
```

```
[1] 80.95077
```

```
# Percentage for EM
em_total / overall_total * 100
```

```
[1] 12.83843
```

X-ray solved for 80.95% and EM accounts for 12.84% of all PDB structures.

- **Q2:** What proportion of structures in the PDB are protein?

```
# Protein-containing structures = Protein (only) + Protein/Oligosaccharide + Protein/NA
protein_total <- stats["Protein (only)", "Total"] +
  stats["Protein/Oligosaccharide", "Total"] +
  stats["Protein/NA", "Total"]

protein_total / overall_total * 100
```

```
[1] 97.9118
```

97.91% of structures in the PDB contain protein.

- **Q3:** Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 226 structures of HIV-1 protease structures in current PDB.

#save pic in text: ![title](filename #)

3. Visualizing the HIV-1 protease structure

The important role of water

Toggle on the display of all water molecules again.

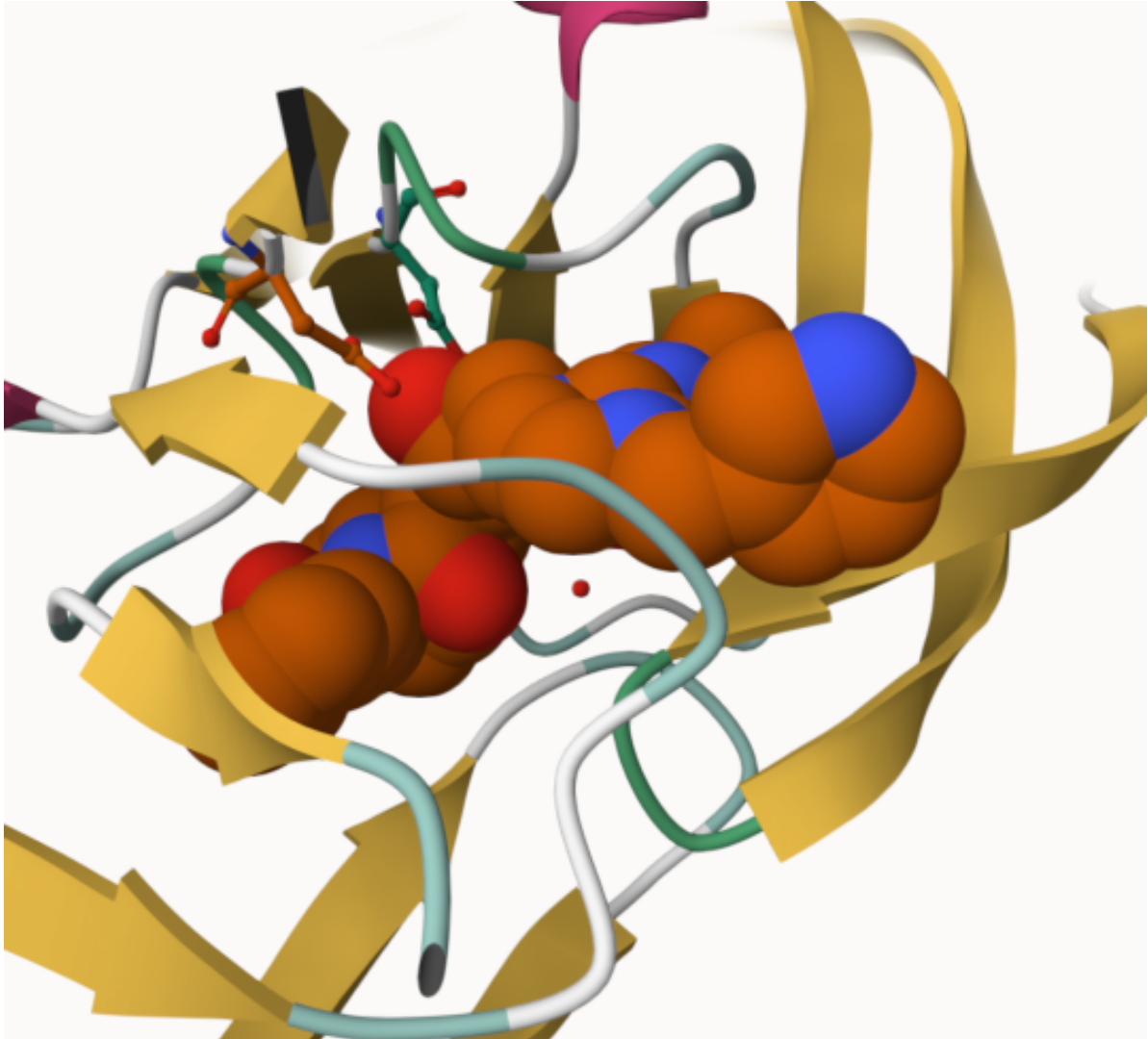
Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

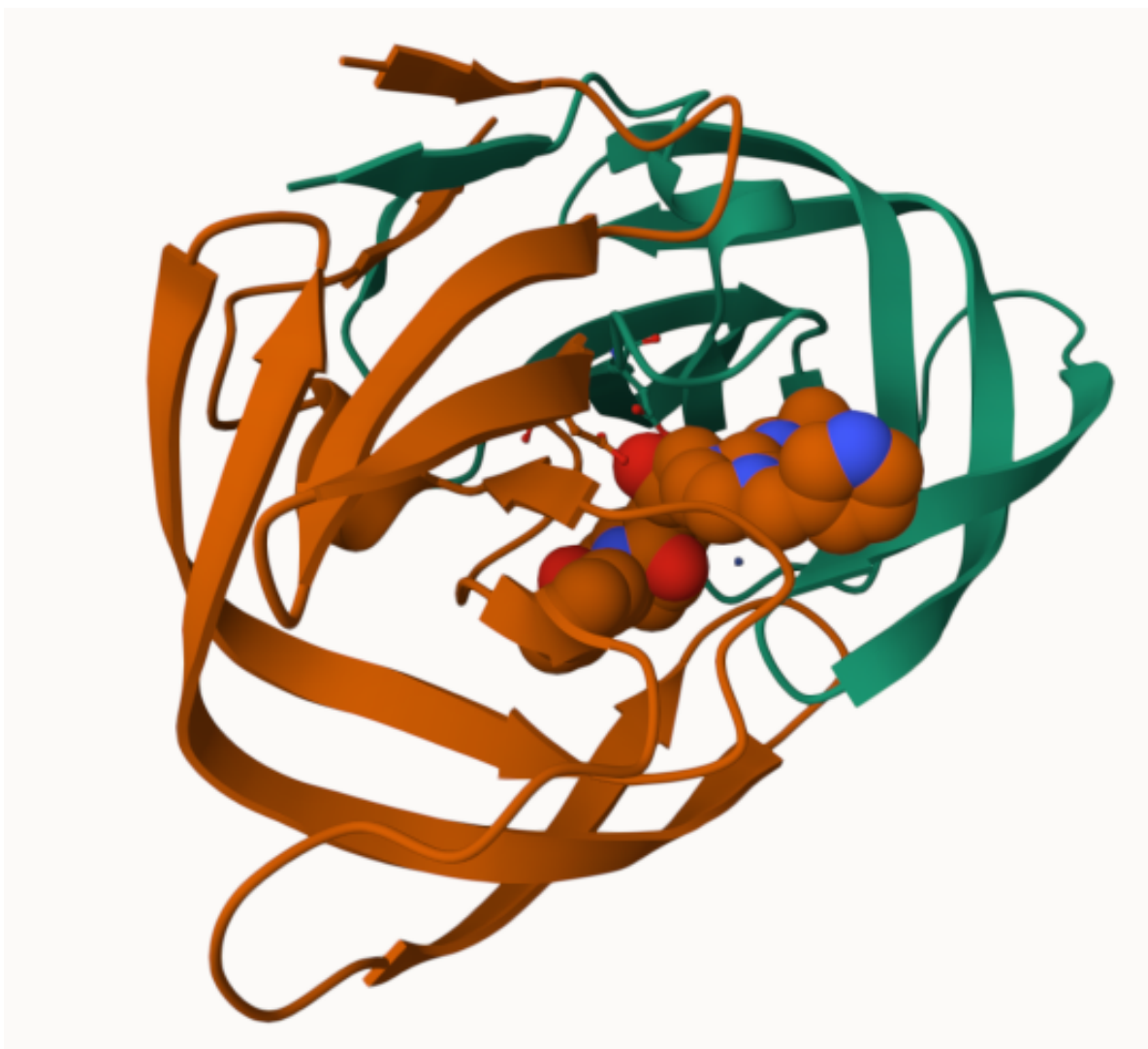
In X-ray crystallography, hydrogen atoms are generally not resolved because they have 1 electrons, making them nearly invisible to X-rays. Since the hydrogen atoms scatter X-rays too weakly to be detected at the 2 Angstrom resolution of this structure, only the oxygen atom of each water molecule is included in the PDB file.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

Yes, the water molecule is HOH 308. This water molecule sits in the binding site and forms a hydrogen-bonding network that bridges the flaps of the protease dimer to the bound inhibitor.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “*Ball & Stick*” for these side-chains). Add this figure to your Quarto document.





Discussion Topic: Can you think of a way in which indinavir, or even larger ligands and substrates, could enter the binding site?

Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

The four-stranded beta-sheet at the interface between the two chains is a key structural element that only forms in the dimer. It can close over the substrate/inhibitor binding site.

4. Introduction to Bio3D in R

- **Q7:** How many amino acid residues are there in this pdb object?

```
# Load the bio3d package
library(bio3d)
```

Warning: package 'bio3d' was built under R version 4.5.3

```
# Read the 1HSG PDB file from the online PDB database
# This downloads the file and parses it into an R object
pdb <- read.pdb("1HSG.pdb")
```

```
# Print a summary of the PDB object
pdb
```

Call: read.pdb(file = "1HSG.pdb")

```
Total Models#: 1
  Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

There are 198 amino acid residues.

- **Q8:** Name one of the two non-protein residues?

The two non-protein residue types are HOH, water, 127 molecules, and MK1, 1 molecule.

- **Q9:** How many protein chains are in this structure?

There are 2 protein chains (A and B), as indicated by “Chains#: 2 (values: A B)”.

```
# View the attributes of the pdb object
attributes(pdb)
```

```
$names
[1] "atom"    "xyz"      "seqres"   "helix"    "sheet"    "calpha"   "remark"   "call"

$class
[1] "pdb" "sse"
```

```
# Access the atom data
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

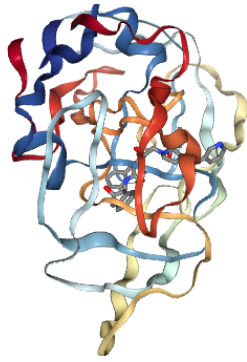
Quick PDB visualization in R

```
# Install the visualization packages (run in Console)
# install.packages("remotes")
# remotes::install_github("bioboot/bio3dview")
# install.packages("NGLVieweR")
```

```
# Load the packages
library(bio3dview)
library(NGLVieweR)
```

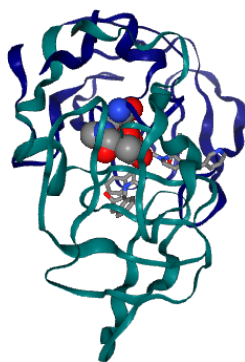
Warning: package 'NGLVieweR' was built under R version 4.5.3

```
# Basic spinning view of the structure
view.pdb(pdb) |>
  setSpin()
```

```
# Custom view: color chains differently and highlight the catalytic Asp 25
sele <- atom.select(pdb, resno = 25)

view.pdb(pdb, cols = c("navy", "teal"),
         highlight = sele,
         highlight.style = "spacefill") |>
setRock()
```



Predicting functional motions of a single structure

```
# Read the Adenylate Kinase structure
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
# Print summary
adk
```

Call: read.pdb(file = "6s36")

Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

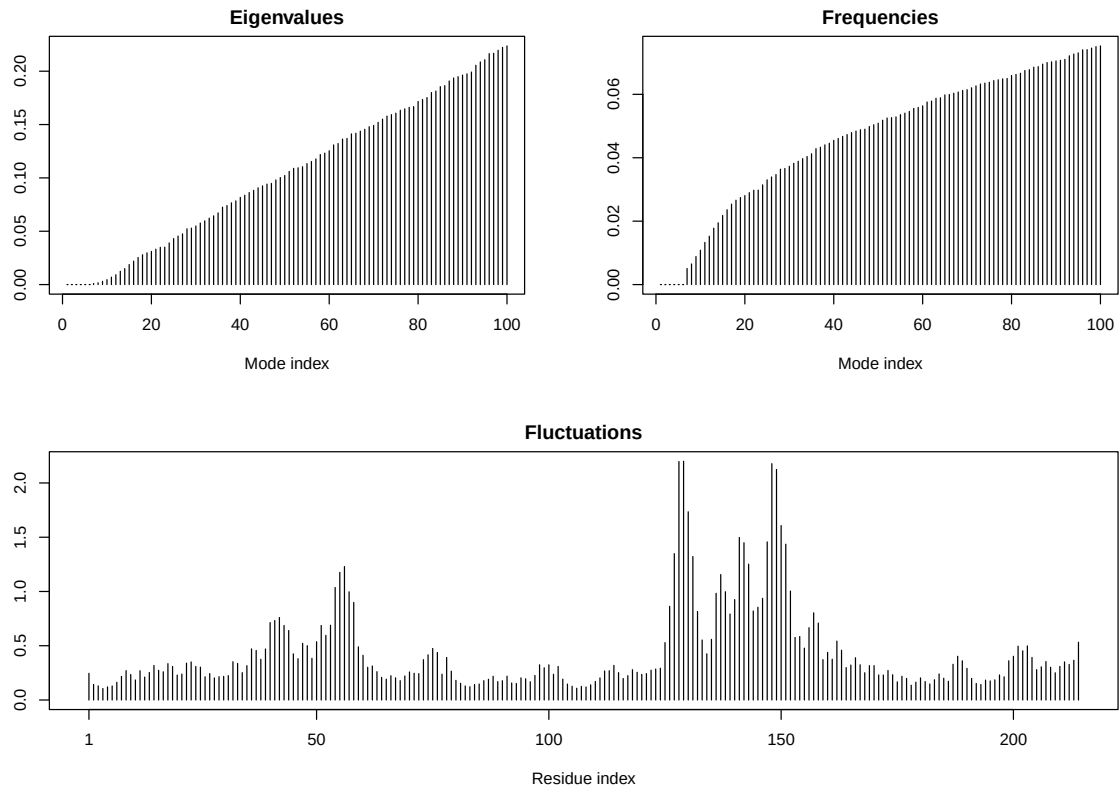
+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

```
# Perform Normal Mode Analysis (NMA) to predict flexibility
# NMA uses an elastic network model to calculate the natural vibrations of the protein
m <- nma(adk)
```

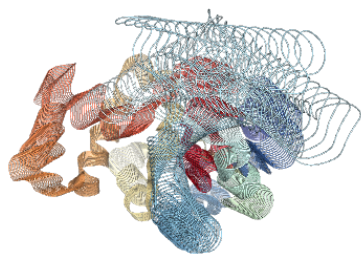
Building Hessian... Done in 0.04 seconds.
Diagonalizing Hessian... Done in 0.56 seconds.

```
# Generate a trajectory PDB file to visualize the predicted motion
# This creates a multi-model PDB file showing the protein "breathing"
mktrj(m, file = "adk_m7.pdb")
```

```
par(mar=c(4, 4, 2, 2)) # bottom, left, top, right margins
# Plot the NMA results
# This shows eigenvalues, frequencies, and per-residue fluctuations
plot(m)
```



```
# Or view directly in R using bio3dview
view.nma(m, pdb = adk)
```



5. Comparative structure analysis of Adenylate Kinase

- **Q10.** Which of the packages above is found only on BioConductor and not CRAN?

msa (Multiple Sequence Alignment). It must be installed via BiocManager::install("msa") rather than the standard install.packages().

- **Q11.** Which of the above packages is not found on BioConductor or CRAN?:

bio3dview. It is installed from GitHub using remotes::install_github("bio-boot/bio3dview")

- **Q12.** True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket? TRUE FALSE

TRUE. The pak package is a modern package manager for R that can install packages from CRAN, Bioconductor, GitHub, GitLab, and BitBucket.

Search and retrieve ADK structures

```
library(bio3d)

# Fetch the sequence for chain A of PDB 1AKE (Adenylate Kinase from E. coli)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

- **Q13.** How many amino acids are in this sequence, i.e. how long is this sequence?

```
# View the sequence
aa
```

```

      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRPTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120
```

```

      121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214

```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

View the sequence

The sequence is 214 amino acids long, as shown by “214 position columns (214 non-gap, 0 gap)” in the output.

may time out:

```

# Blast search the PDB database to find related structures
# This may time out due to server load - see the backup below
# b <- blast.pdb(aa)

# Plot and filter results
# hits <- plot(b)

# List top hits
# head(hits$ pdb.id)

# use these pre-identified PDB IDs
hits <- NULL
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A',
                 '5EJE_A', '1E4Y_A', '3X2S_A', '6HAP_A', '6HAM_A',
                 '4K46_A', '3GMT_A', '4PZL_A')

```



```
# Download the PDB files for all hits
# split=TRUE extracts individual chains; gzip=TRUE compresses files
files <- get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb exists. Skipping download
```

```
|
|
| 0%
|====
| 8%
|=====
| 15%
|=====
| 23%
|=====
| 31%
|=====
| 38%
|=====
| 46%
|=====
| 54%
|=====
| 62%
|=====
| 69%
|=====
| 77%
|=====
| 85%
|=====
| 92%
|=====
| 100%
```

```
# Align the structures using multiple sequence alignment
# fit=TRUE superimposes all structures onto a common reference frame
# exefile="msa" tells bio3d to use the Bioconductor msa package
pdbs <- pdbaln(files, fit = TRUE, exefile = "msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
```

```

pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.    PDB has ALT records, taking A only, rm.alt=TRUE
...

```

Extracting sequences

```

pdb/seq: 1    name: pdbs/split_chain/1AKE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2    name: pdbs/split_chain/6S36_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3    name: pdbs/split_chain/6RZE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4    name: pdbs/split_chain/3HPR_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5    name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6    name: pdbs/split_chain/5EJE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7    name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8    name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9    name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10   name: pdbs/split_chain/6HAM_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11   name: pdbs/split_chain/4K46_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13   name: pdbs/split_chain/4PZL_A.pdb

```

^^ upper step done=

- (1) aligns the amino acid sequences to identify equivalent positions
- (2) superimposes the 3D coordinates so all structures overlap as much as possible. This is essential for comparing conformations.

Annotate collected PDB structures

```
# Get annotation info (species, method, resolution, etc.) from PDB
ids <- basename.pdb(pdbids$id)
anno <- pdb.annotate(ids)

# See which species these structures come from
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
# View all annotation data
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-ray
6S36_A	6S36	A	Protein	214	X-ray
6RZE_A	6RZE	A	Protein	214	X-ray
3HPR_A	3HPR	A	Protein	214	X-ray
1E4V_A	1E4V	A	Protein	214	X-ray
5EJE_A	5EJE	A	Protein	214	X-ray
1E4Y_A	1E4Y	A	Protein	214	X-ray
3X2S_A	3X2S	A	Protein	214	X-ray

6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					
3GMT_A	3GMT	A	Protein	230	X-
ray					
4PZL_A	4PZL	A	Protein	242	X-
ray					

	resolution	scopDomain	pfam
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
3X2S_A	2.80	<NA>	<NA>
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)
3GMT_A	2.10	<NA>	<NA>
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)

	ligandId
1AKE_A	AP5
6S36_A	CL (3),NA,MG (2)
6RZE_A	NA (3),CL (2)
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,CO
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG
6HAP_A	AP5
6HAM_A	AP5
4K46_A	ADP,AMP,PO4
3GMT_A	SO4 (2)
4PZL_A	GOL,FMT,CA

	ligandName
1AKE_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6S36_A	CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A	SODIUM ION (3),CHLORIDE ION (2)

3HPR_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
1E4V_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
5EJE_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2),	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
6HAM_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
4K46_A	ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION	
3GMT_A		SULFATE ION (2)
4PZL_A		GLYCEROL,FORMIC ACID,CALCIUM ION
	source	
1AKE_A	Escherichia coli	
6S36_A	Escherichia coli	
6RZE_A	Escherichia coli	
3HPR_A	Escherichia coli K-12	
1E4V_A	Escherichia coli	
5EJE_A	Escherichia coli 0139:H28 str. E24377A	
1E4Y_A	Escherichia coli	
3X2S_A	Escherichia coli str. K-12 substr. MDS42	
6HAP_A	Escherichia coli 0139:H28 str. E24377A	
6HAM_A	Escherichia coli K-12	
4K46_A	Photobacterium profundum	
3GMT_A	Burkholderia pseudomallei 1710b	
4PZL_A	Francisella tularensis subsp. tularensis SCHU S4	
1AKE_A	STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIB.	
6S36_A		
6RZE_A		
3HPR_A		
1E4V_A		
loop		
5EJE_A		Crys
1E4Y_A		
loop		
3X2S_A		
conjugated adenylate kinase		
6HAP_A		
6HAM_A		

4K46_A
3GMT_A
4PZL_A

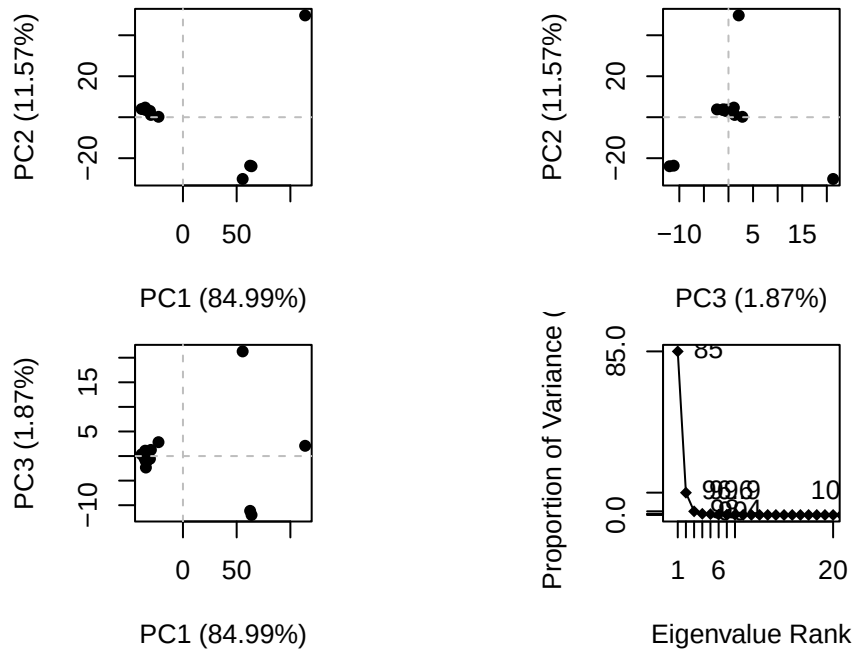
The crys

		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al.	J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al.	Biochemistry (2019)	0.16320	0.23560
6RZE_A	Rogne, P., et al.	Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al.	Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al.	Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al.	Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al.	Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al.	Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al.	To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al.	Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al.	To be published	0.19360	0.23680
	rWork	spaceGroup		
1AKE_A	0.19600	P 21 2 21		
6S36_A	0.15940	C 1 2 1		
6RZE_A	0.18190	C 1 2 1		
3HPR_A	0.20620	P 21 21 2		
1E4V_A	0.19600	P 21 2 21		
5EJE_A	0.18630	P 21 2 21		
1E4Y_A	0.17800	P 1 21 1		
3X2S_A	0.20700	P 21 21 21		
6HAP_A	0.22370	I 2 2 2		
6HAM_A	0.20311	P 43		
4K46_A	0.16730	P 21 21 21		
3GMT_A	0.23500	P 1 21 1		
4PZL_A	0.19130	P 32		

Principal Component Analysis (PCA)

```
# Perform PCA on the aligned structure ensemble
# PCA finds the directions of greatest structural variation
pc.xray <- pca(pdbx)

# Plot PCA results: shows PC1 vs PC2, PC3 vs PC2, and a scree plot
plot(pc.xray)
```



RMSD clustering and PCA visualization

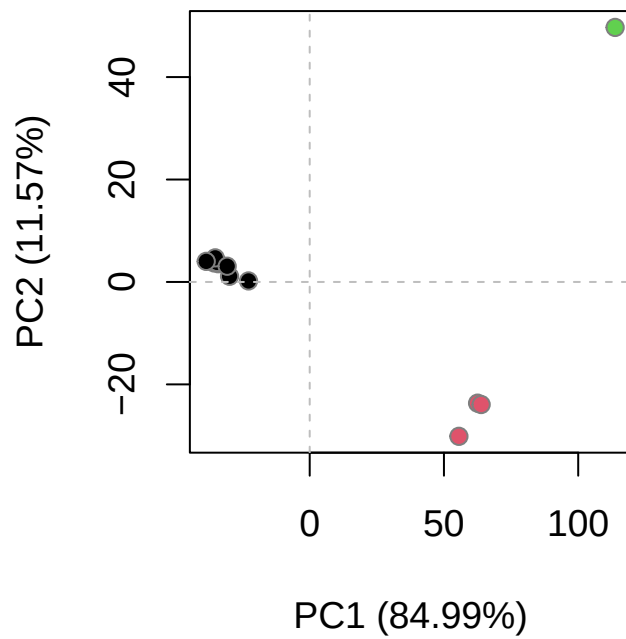
```
# Calculate all pairwise RMSD values between structures
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 204 non NA positions

```
# Hierarchical clustering based on RMSD
hc.rd <- hclust(dist(rd))

# Cut into 3 groups
grps.rd <- cutree(hc.rd, k = 3)

# Plot PC1 vs PC2, colored by RMSD-based cluster
plot(pc.xray, 1:2, col = "grey50", bg = grps.rd, pch = 21, cex = 1)
```

PCA visualization

```
# Generate a trajectory along PC1 to visualize the dominant motion
pc1 <- mktrj(pc.xray, pc = 1, file = "pc_1.pdb")
```

```
library(ggplot2)
```

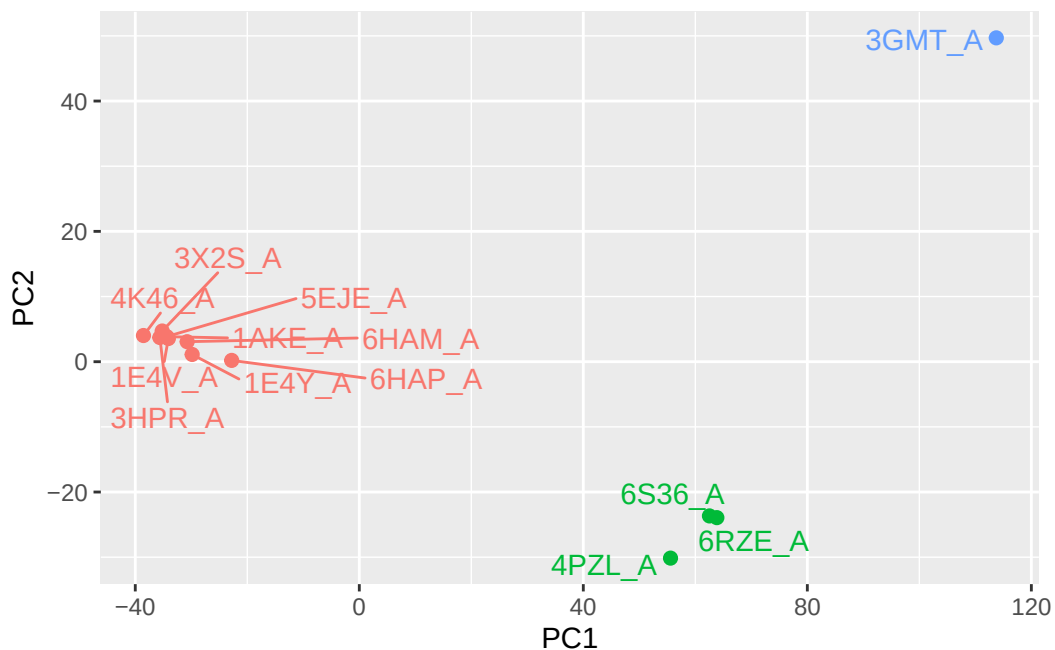
Warning: package 'ggplot2' was built under R version 4.5.3

```
library(ggrepel)
```

Warning: package 'ggrepel' was built under R version 4.5.3

```
# Create a dataframe for ggplot
df <- data.frame(
  PC1 = pc.xray$z[, 1],
  PC2 = pc.xray$z[, 2],
  col = as.factor(grps.rd),
  ids = ids
)
```

```
# Create the plot with labeled points
p <- ggplot(df) +
  aes(PC1, PC2, col = col, label = ids) +
  geom_point(size = 2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```



6. Normal mode analysis [optional]

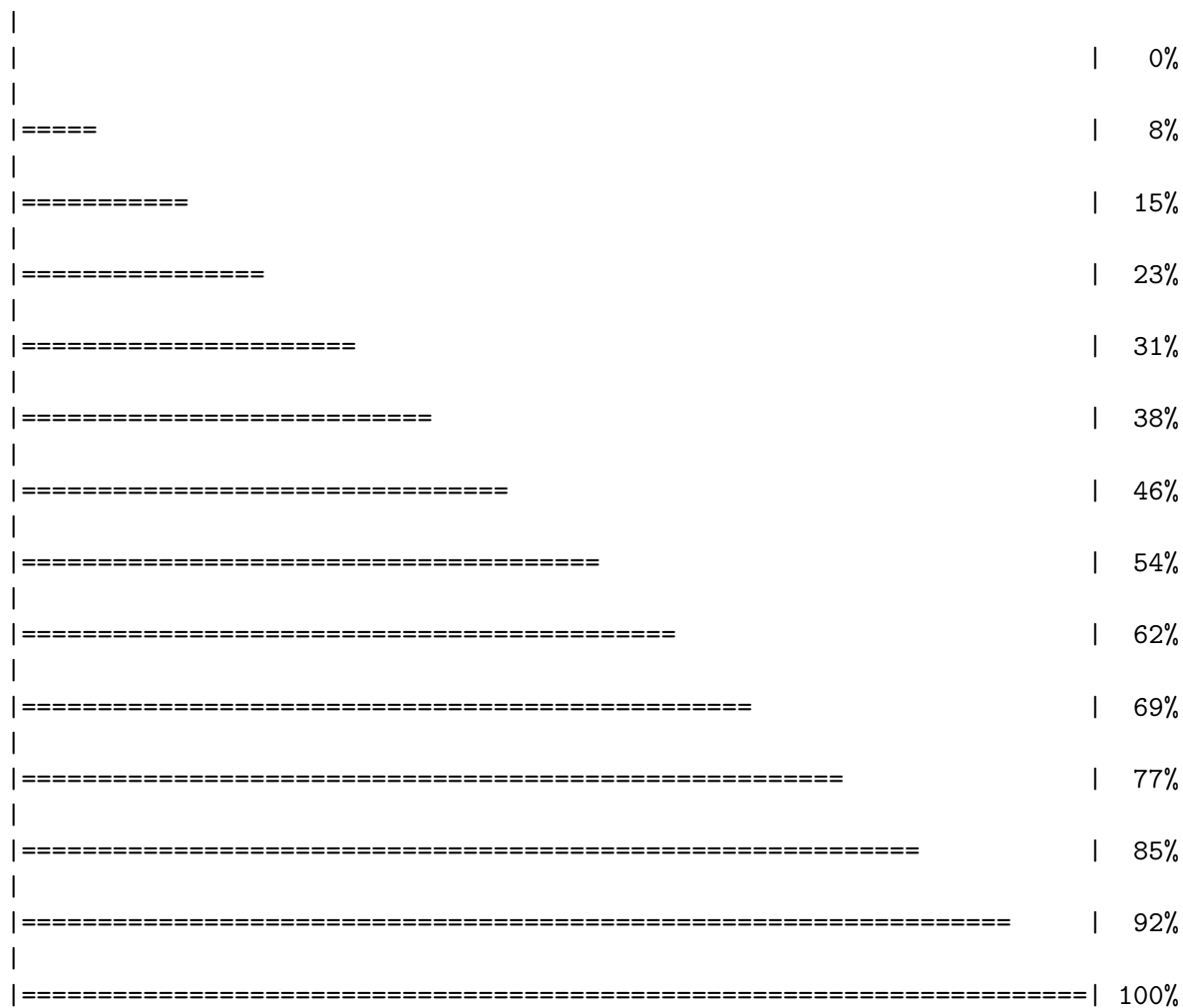
- **Q14;** What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

```
# Perform NMA on ALL structures in the ensemble
modes <- nma(pdbbs)
```

Details of Scheduled Calculation:

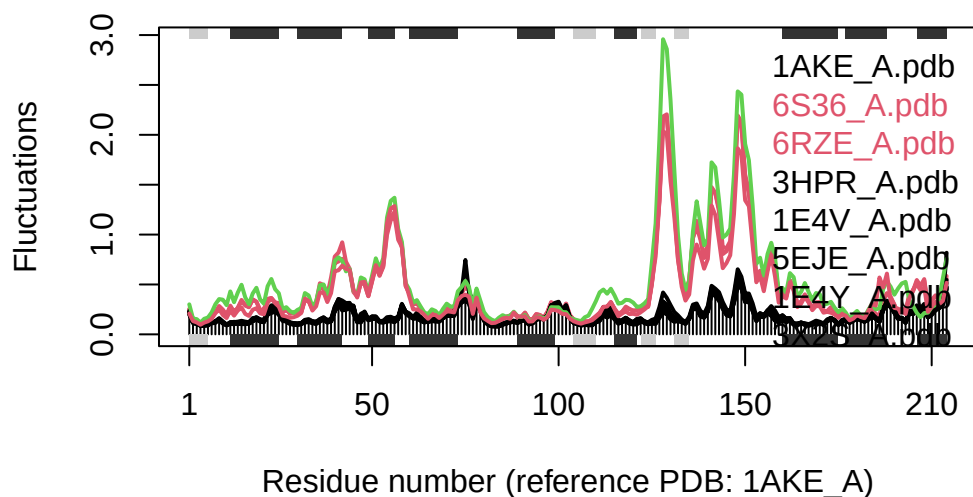
- ... 13 input structures
- ... storing 606 eigenvectors for each structure

```
... dimension of x$U.subspace: ( 612x606x13 )
... coordinate superposition prior to NM calculation
... aligned eigenvectors (gap containing positions removed)
... estimated memory usage of final 'eNMA' object: 36.9 Mb
```



```
# Plot fluctuation profiles, colored by RMSD cluster
plot(modes, pdb, col = grps.rd)
```

Extracting SSE from pdb\$sse attribute



The black lines (ligand-bound group) and the colored lines (ligand-free group) show similar overall fluctuation patterns. It means the same residues tend to be flexible or rigid across all structures. Yet, they differ most in two regions: around residues 30-65 (the “lid” domain) and residues 125-165 (the “NMP-binding” domain). In the open conformation, these regions show larger fluctuations, while in the black lines, these same regions are more rigid because nucleotide binding stabilizes them. These are the two mobile domains that clamp down on the substrates during catalysis.